

Data Quality Checks for Data Engineers

Chapter 1: Null or Missing Values Check

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

Discussion: Explain the business rule, expected output, production considerations, interview tips, and monitoring metrics. Replace the placeholder SQL and Spark snippets with project-specific implementations.

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Chapter 2: Primary Key Uniqueness Check

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 3: Duplicate Record Detection

Purpose: Ensure data meets expected quality standards before downstream

consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 4: Referential Integrity Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 5: Data Type Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 6: Numeric Range Checks

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 7: String Length Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 8: Regex Pattern Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 9: Allowed Values/Domain Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 10: Business Rule Consistency

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

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SELECT * FROM table WHERE <validation_condition>;
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Sample PySpark

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Best Practice: Log failed records into a quarantine table with rejection reason and audit

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Chapter 11: Cross-Column Consistency Checks

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 12: Timeliness/Freshness Check

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 13: Completeness Check

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 14: Volume Checks

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

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SELECT * FROM table WHERE <validation_condition>;
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Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 15: Distribution Checks

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

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SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

Discussion: Explain the business rule, expected output, production considerations, interview tips, and monitoring metrics. Replace the placeholder SQL and Spark snippets with project-specific implementations.

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Chapter 16: Outlier Detection

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 17: Schema Drift Detection

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 18: Duplicate File Ingestion

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

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SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 19: Negative Value Checks

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 20: Percentage/Total Consistency Checks

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 21: Hierarchy Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

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SELECT * FROM table WHERE <validation_condition>;
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Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 22: Audit Column Consistency

Checks **Purpose:** Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 23: Foreign Key Existence Check

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
```

Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 24: Date Validity Check

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
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Sample PySpark

```
df.filter(<validation_condition>)
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Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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Chapter 25: PII Masking Validation

Purpose: Ensure data meets expected quality standards before downstream consumption.

Example: In a healthcare patient table, validate the relevant rule before loading into the Silver layer.

Sample SQL

```
SELECT * FROM table WHERE <validation_condition>;
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Sample PySpark

```
df.filter(<validation_condition>)
```

Best Practice: Log failed records into a quarantine table with rejection reason and audit metadata.

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